

Algorithm-Based Tutoring Across Borders: How Gauth and Doubao Aixue Reflect and Reinforce U.S. and Chinese Educational Systems

Introduction

ByteDance has developed two similar AI-powered homework help and tutoring platforms, Gauthmath in the United States and 豆包爱学 (Doubao Aixue) in mainland China. Both use AI to support learners with homework help and step-by-step explanations, covering K12 and early college-level knowledge. They belong to what Reich (2020) calls the algorithm-based genre of educational technology: systems that take student inputs (photos of problems, text prompts, taps, and clicks) and use algorithms to generate step-by-step solutions, feedback, and personalized practice, rather than simply delivering static content or hosting complete online courses. In this paper, I compare how ByteDance's two algorithm-based AI math tutors have been shaped by the very different educational systems of the United States and China. I argue that Gauth, operating in a decentralized, market-driven U.S. context, functions mainly as a consumer-facing homework accelerator, while Doubao Aixue, operating in a centralized, exam-driven, and policy-steered Chinese context, positions itself as a policy-compatible AI teacher-companion.

Across three parts, I first situate both platforms within their policy, cultural, and social environments (Part I), then analyze their features through the theoretical lenses of instructionism and constructionism pedagogies (Part II), and finally draw on Reich's concepts of the "curse of the familiar" and the EdTech Matthew Effect to argue that despite their technical innovation, these platforms may ultimately reinforce rather than disrupt the core logics and inequalities of the education systems in which they operate (Part III).

Part I. Policy, Cultural, and Social Foundations of AI Tutoring: A Comparative Analysis of Gauth (U.S.) and Doubao Aixue (China)

Gauth and Doubao Aixue are not just two versions of the same product translated for different markets. Instead, they reflect very different educational philosophies and policy systems in the United States and China. These platforms show how AI in education gets reshaped by local ideas about school, authority, and what “good learning” looks like. In the U.S., Gauth operates as a

consumer-oriented homework helper. In China, Doubao functions more like a policy-aligned AI teacher. These are not just design choices; they reflect the governing structures and social pressures around each product.

The U.S. and China run their education systems very differently. In the United States, the Every Student Succeeds Act (ESSA), signed into law in 2015, sets national expectations for equity, standards, and accountability while also emphasizing state and local control over how those expectations are implemented (U.S. Department of Education, n.d.). ESSA requires states to administer yearly assessments and intervene in struggling schools, but it leaves a lot of room for states, districts, and individual schools to design curricula and choose instructional materials (U.S. Department of Education, n.d.). There is no single national curriculum and no required digital platform for homework help. Students, families, and sometimes schools decide whether to use tools like Gauth—the federal government does not make those choices for them.

In many U.S. secondary and postsecondary settings, performance indicators like GPAs, standardized test scores, and college admissions outcomes play an important role in shaping students' and families' decisions. Families with more financial and cultural resources are more likely to invest in private tutoring, test-prep services, and educational technologies as part of broader strategies to support academic achievement. As Boudreau (2021) reports, “during the pandemic, the industry saw a boom, with 1 in 5 upper-income families hiring someone to aid with their child’s education,” highlighting how additional academic support is especially accessible to families with higher incomes. In this landscape, tools like Gauth are not arriving in a vacuum; they are often layered on top of existing advantages, becoming one more resource that some families can stack alongside private tutoring and strong neighborhood schools. Gauth enters this landscape as a direct-to-consumer app. Individual users download it from app stores; it competes with other homework tools, and it operates on a free and subscription model. Its survival depends on attracting and keeping users in an open marketplace, not on approval from a centralized ministry.

This commercial setting helps explain why Gauth’s design emphasizes speed, convenience, and task efficiency. Its core features: photo-based problem solving, step-by-step solutions, reading summaries, and writing assistance, position AI as a service that helps students complete

academic work faster. Tech reporting describes Gauth (originally Gauthmath) as one of the top education apps in the U.S. market, with tens of millions of downloads and a large global user base (Schuler, 2025). The app lets students snap photos of math problems and receive worked solutions, and it uses a large pool of human tutors to handle harder questions (Schuler, 2025). User reviews also emphasize that when the AI struggles or makes mistakes, students can escalate problems to human experts through 24/7 tutor support, creating a hybrid model in which algorithmic solutions are backed by always-available human help (Amdad, 2024). But such unlimited detailed explanations and faster tutor access come with the paid “Gauth Plus” plan (Schuler, 2025). In this context, Gauth makes sense as a tool that promises: “Use this app, and you can get through your homework more quickly, understand the steps, and maybe perform better on quizzes and tests.”

Doubao Aixue operates in a very different environment. China’s national education system is organized around a unified curriculum, high-stakes exams, and strong control from the Ministry of Education and other central bodies. In July 2021, the General Offices of the Chinese Communist Party Central Committee and the State Council released the *Opinions on Further Reducing the Burden of Homework and Off-Campus Training for Compulsory Education Students*, commonly known as the “Double Reduction” (双减) policy (Chen & Lin, 2024). This policy targeted two main sources of what Chinese researchers call students’ “academic burden” (负担, *fudan*): heavy homework inside school and the rapid expansion of off-campus, exam-focused “shadow education” (Chen & Lin, 2024).

Here, academic burden is not just “too much homework.” It is tied to deeper cultural and social factors: Confucian values that strongly emphasize educational achievement, an exam-based system centered on the high-stakes *gaokao* college entrance exam, and parental beliefs that academic success is the main path to social mobility, family pride, and respect (Chen & Lin, 2024). For decades, national policies have tried to reduce this burden, but families and schools have continued to push students toward long study hours and extra tutoring.

Chen and Lin (2024) suggest that Double Reduction is part of a larger policy shift from narrow exam orientation toward “quality education” (素质教育, *suzhi jiaoyu*). Quality education aims not only at high test scores but also at students’ moral, intellectual, physical, aesthetic, and labor

development—often described as “five educations in integration” (五育融合, *wuyuronghe*). The policy’s core goals include reducing excessive homework and off-campus training, improving the quality of teaching in regular schools so that normal school education is “enough,” and easing inequality by limiting the impact of family income and private tutoring on student outcomes (Chen & Lin, 2024). To support these aims, Double Reduction sets strict limits on daily homework time and sharply restricts for-profit subject-based tutoring in core academic subjects.

Official statistics and early studies show some positive effects after Double Reduction, such as less reported homework, more time for extracurricular activities, and a dramatic drop in the number of licensed subject-based tutoring institutions (Chen & Lin, 2024). However, both research and on-the-ground reporting suggest that the deeper exam pressure has not gone away. Nguyen and Feng (n.d.), for example, describe how extra academic lessons have shifted from private tutoring centers into public schools. Many high schools now organize long after-school “self-study” sessions, weekend tests, and additional practice classes under teacher supervision. University students in their article report still studying late into the night and taking exams even on holidays; one student remarks that “nothing really changed” for them in terms of pressure (Nguyen & Feng, n.d.).

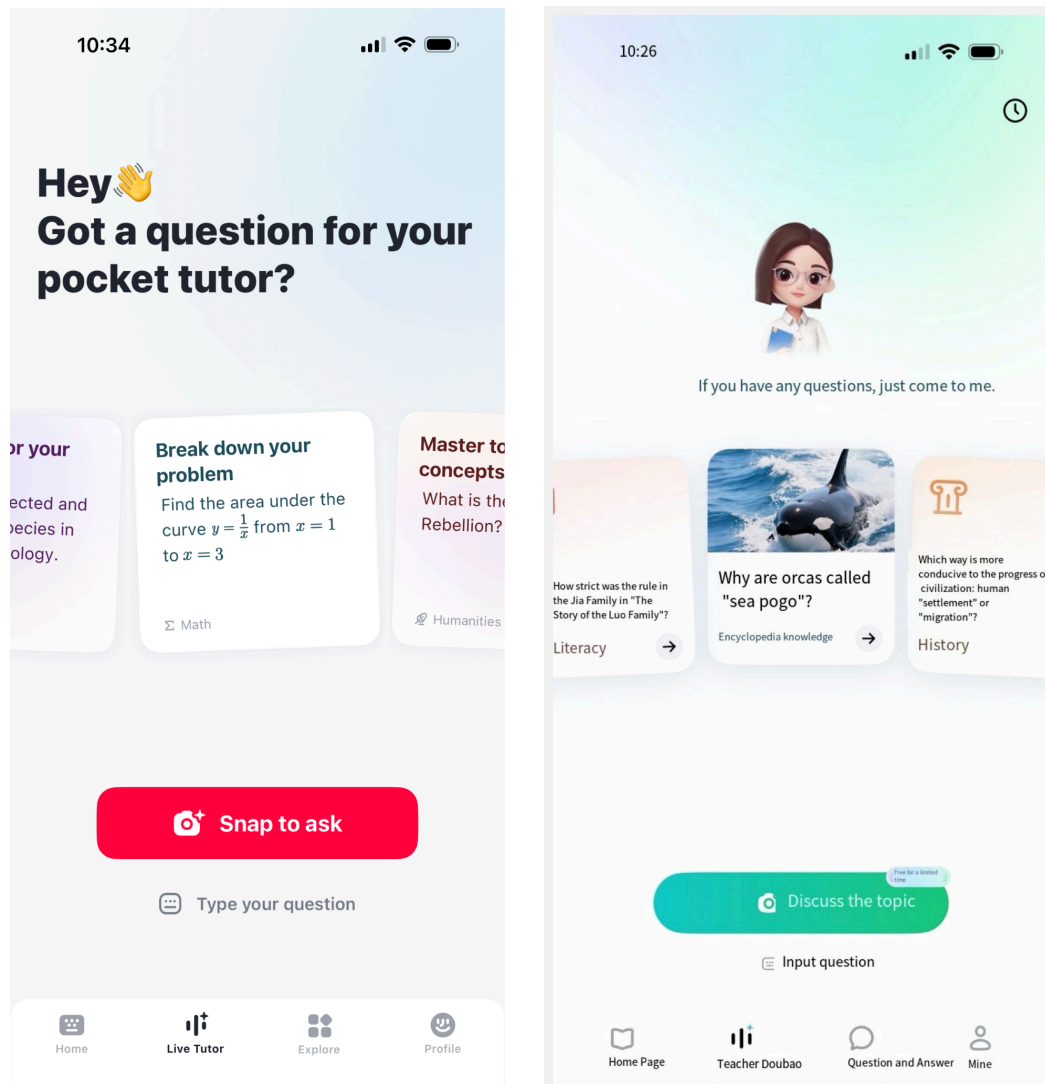
Students and teachers often use the term *neijuan* (内卷), or “involution,” to describe this situation: a feeling of being stuck in endless, cutthroat competition where everyone must keep working harder just to avoid falling behind. Under these conditions, families still look for ways to give their children an edge, even if formal private tutoring is banned (Nguyen & Feng, n.d.). Some extra teaching moves underground or is relabeled as “self-study,” and schools in wealthier urban areas are often better able to provide strong after-school services than those in rural or under-resourced areas (Chen & Lin, 2024). This means the policy’s goals of reducing both academic burden and inequality are hard to fully achieve in practice.

Doubao Aixue appears inside this post–Double Reduction landscape. It does not simply enter a free consumer market; it enters a tightly managed policy environment where educational technologies are expected to match Double Reduction’s values. Doubao cannot just respond to what users want; it must also respect national curriculum standards, align with official ideas

about “quality education,” and follow strict rules against recreating banned forms of commercial test prep. As a result, Doubao functions less like a purely market-driven product and more like a state-compatible educational infrastructure. Its features must meet a double test: they have to support students’ needs for homework help and practice, and they also have to show that the platform is giving “proper” guidance, reinforcing official content, and supporting student well-being rather than simply rebuilding the old shadow education system (Chen & Lin, 2024; Nguyen & Feng, n.d.).

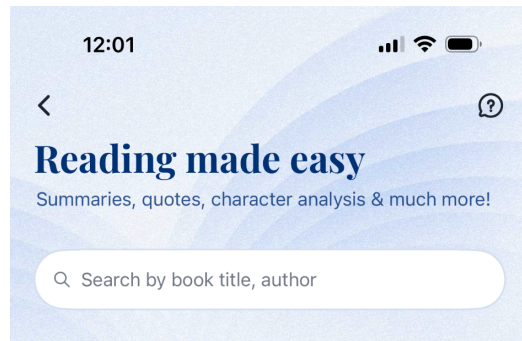
These different policy regimes create distinct relationships between learners and platforms. Gauth operates mainly within a consumer marketplace where students and families make individual choices about academic support tools. Its value proposition is practical and efficiency-focused: the app helps users solve problems more quickly, clarify complex steps, and complete assignments with less friction. Learning is framed as an individual pursuit in a performance-oriented setting, and the platform’s success rests on whether users find it genuinely helpful for their schoolwork. Doubao Aixue positions learners within a different framework: one where family expectations and state education policy are both very present. The platform presents itself not just as a problem-solving tool but as a trustworthy AI companion who provides homework guidance, explains ideas step by step, and listens to students’ feelings. Learning appears as both an intellectual and developmental process, linked to steady effort, “good” study habits, and alignment with national curricular goals.

These differences are visible in how the platforms organize core academic support. Gauth’s interface centers on fast problem submission and immediate solution delivery: you upload a photo of a math or science question and get a complete worked answer, with AI acting as an efficient problem-solving co-pilot. Doubao Aixue offers a similar photo-based Q&A feature but places it inside a wider set of teacher-like tools, including homework grading and detailed explanations. Instead of presenting itself as a neutral tool, the platform adopts an instructional persona, explicitly taking on the “teacher” role in both text and visuals. Explanations sound like classroom teaching and often include caring or encouraging phrases. This design aligns with policy expectations that educational technologies should guide students in the “right” way rather than serve as shortcuts that weaken the authority of schools and teachers.



[Left: Gauth; Right: Doubao Aixue; Youdao Translator translates Doubao Aixue's screenshots]

The platforms also differ in how they support writing and creative expression. Gauth's writing tools function mainly as productivity aids for generating or refining academic text. They provide templates for essays, outlines, and revisions, emphasizing proper structure and efficient output. Doubao Aixue offers both composition support and picture-book creation. In the picture-book feature, students suggest a story idea, and the AI helps build a narrative with images. Here, the platform shifts from assessor and explainer to co-creator, giving students more room to express their own ideas. At the same time, this creative mode does not replace exam writing; on the same platform, students also practice tightly structured essays designed to match rubric-based scoring in Chinese and English.



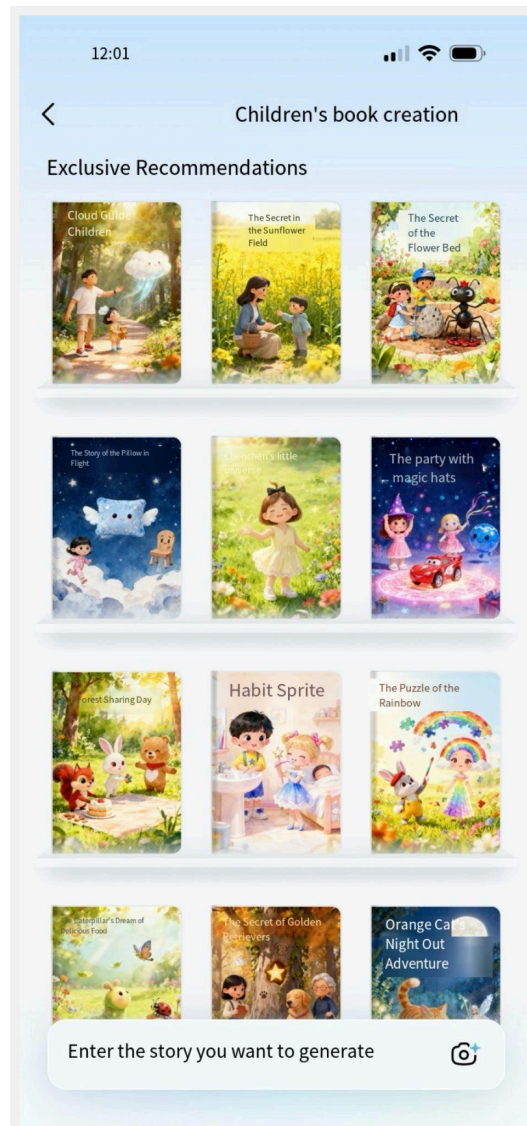
Popular Books



Works of Shakespeare

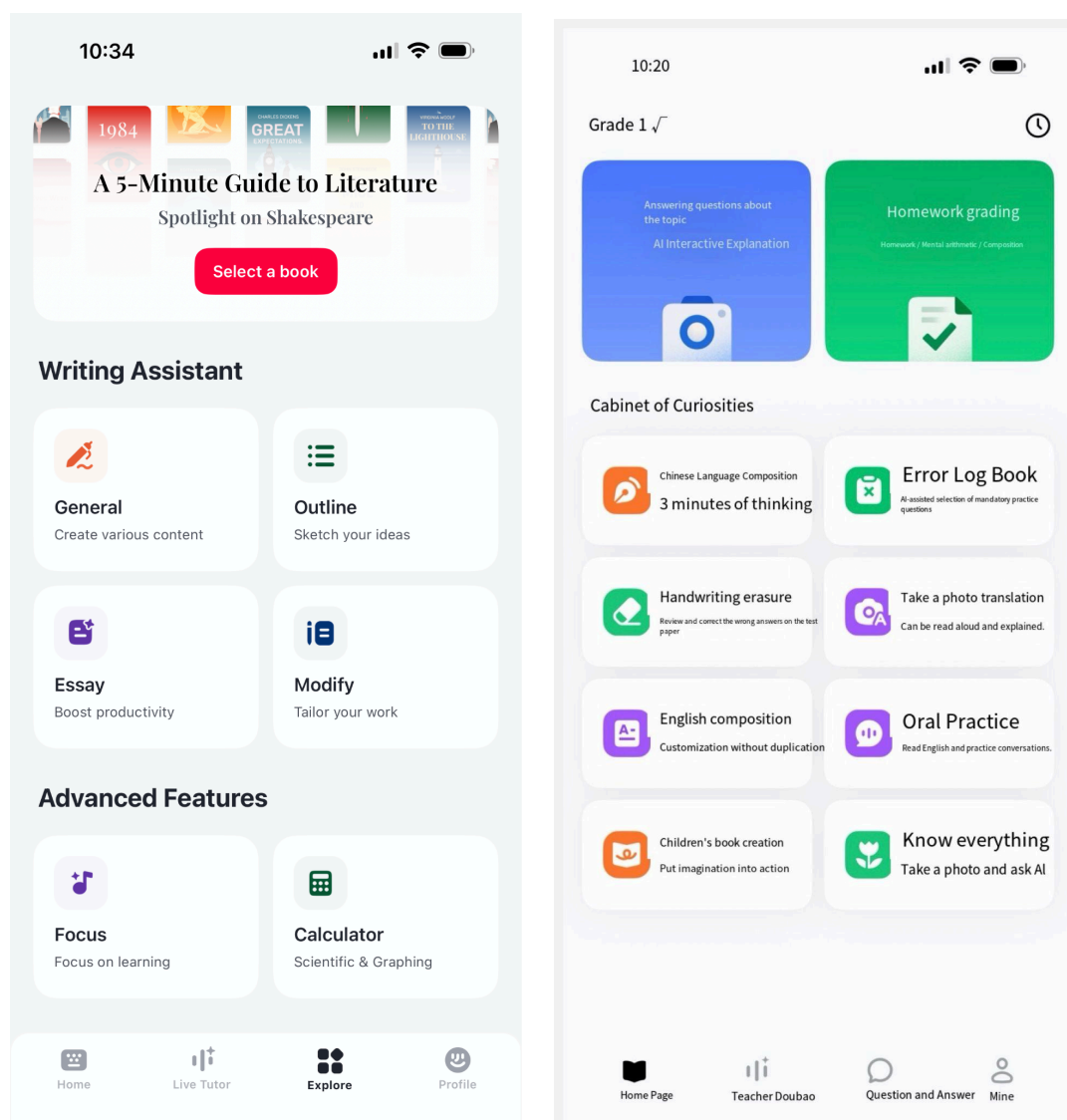
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[Left: Gauth; Right: Doubao Aixue; Youdao Translator translates Doubao Aixue's screenshots]

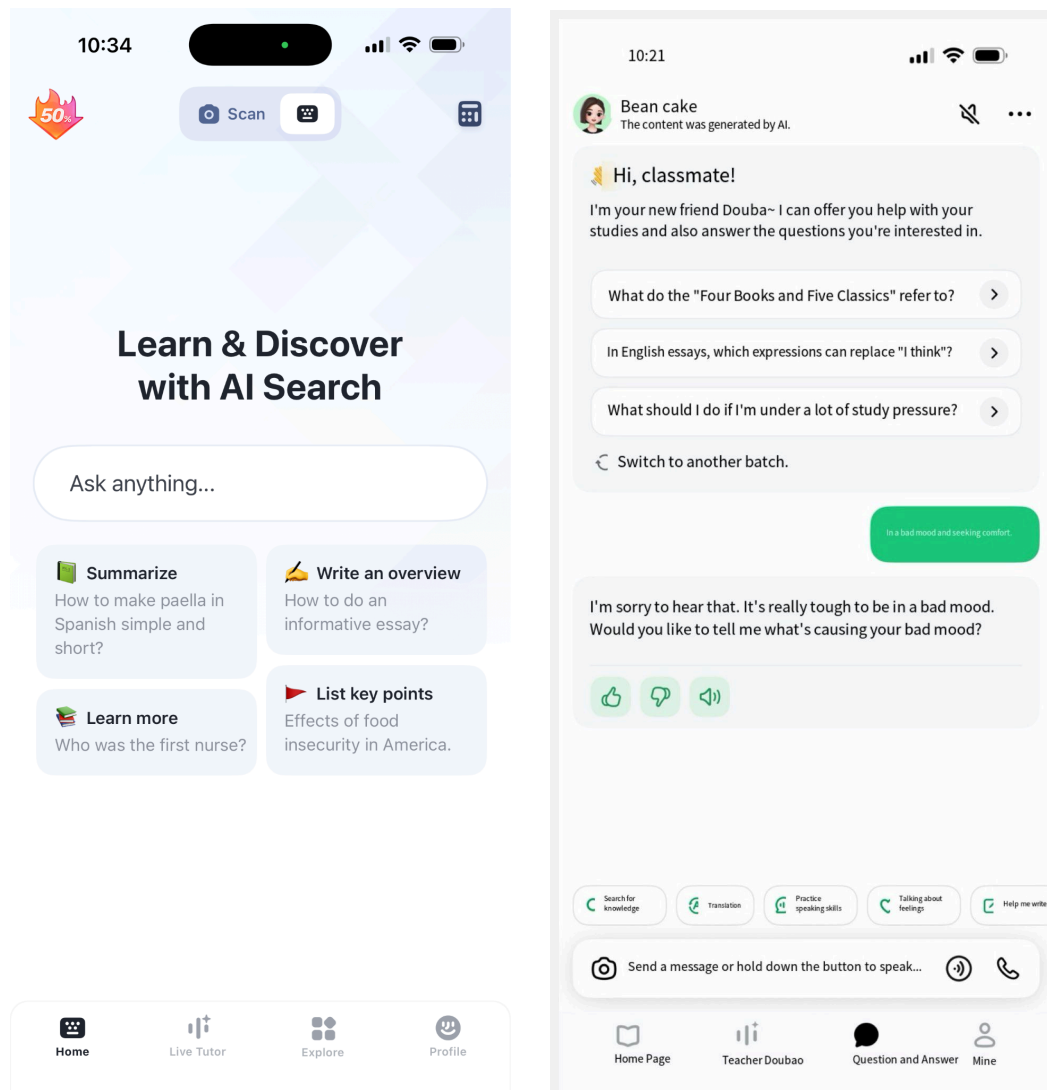
The most telling difference lies in how the platforms handle error, practice, and mastery. In Doubao, the error notebook formalizes a long-standing Chinese study habit: writing down mistakes, revisiting them, and slowly eliminating them through repeated practice. By automatically collecting and organizing errors, the platform turns this familiar paper-based method into a digital, data-driven system. Students can see patterns in their mistakes, return to complex topics, and track progress over time. Gauth allows repeated attempts and step-by-step explanations, but it does not log errors in the same ritualized way. Its primary focus is on solving the current problem rather than building a long-term record of misunderstandings.



[Left: Gauth; Right: Doubao Aixue; Youdao Translator translates Doubao Aixue's screenshots]

Finally, the platforms diverge in how they use AI beyond strictly academic work. Gauth extends into general knowledge search and study support, but it mostly frames these tools in informational and task-oriented terms. Doubao Aixue goes further into daily life and emotional experience. Features that identify objects in the environment, support spoken language practice, or offer a “mood” space for students to talk about how they feel position the platform as a complete study–life companion. The AI is not only a solver of math problems but also a partner in practicing oral English, answering curiosities, and coping with stress or frustration. This design reflects a Chinese expectation that education is holistic and that tools used by children

and teenagers should support both academic success and emotional regulation under intense pressure.



[Left: Gauth; Right: Doubao Aixue; Youdao Translator translates Doubao Aixue's screenshots]

Taken together, these contrasts show that Gauth and Doubao Aixue are not simply different interface layers on the same technology. They embody two different ideas about automated learning. Gauth is embedded in a consumer marketplace shaped by decentralized governance and user choice; it offers AI as an efficient service for helping with schoolwork. Doubao is embedded in a centralized, exam-driven system undergoing major reforms; it offers AI as a regulated teacher-companion designed to align with state-approved curricula and the idea of “quality

education.” Yet despite these differences, both platforms largely uphold instructionist traditions, positioning learners mainly as recipients of correct knowledge. This shared instructional foundation sets the stage for the theoretical analysis in Part II.

Part II. Instructionism and Constructionism in AI Tutoring Platforms: A Pedagogical Analysis of Gauth and Doubao Aixue

Building on the policy, cultural, and social analysis in Part I, this section looks more closely at how Gauth and Doubao Aixue organize learning at the level of concrete features. At the core of both platforms is a photo-based problem-solving feature: “Snap to Ask”. Pedagogically, these features are strongly instructionist. A learner submits an image of a math or science problem and receives a worked solution with step-by-step procedures. The main cognitive work of selecting a method, structuring the solution, and sequencing the steps is carried out by the system, while the learner’s role is to study and, ideally, imitate the model. This photo-based solution feature functions as an implementation of the worked example effect: instead of solving the problem from scratch, learners study a fully worked solution and then apply the same schema to similar situations (Sweller et al., 1998).

Doubao Aixue’s “AI teacher” mode (Teacher Doubao) extends this instructional logic into a more explicitly teacher-like format. Through voice explanations and dynamic board-writing, the system presents itself as a classroom instructor: it selects the content, structures the explanation, and delivers it in a largely one-way flow. At the same time, this feature may lower barriers for students who do not have ready access to human tutors, showing how an instructionist design can still serve significant equity and accessibility goals.

Doubao’s error notebook introduces a more hybrid pattern. On one level, the error notebook formalizes a widely practiced exam-preparation routine: recording mistakes, revisiting them, and practicing correct forms until they become fluent. In that sense, it supports an instructionist emphasis on error correction and procedural accuracy. Yet by externalizing a learner’s personal error history, the feature also creates opportunities for metacognitive reflection. Students can begin to see patterns in the types of tasks they miss, re-experience their own problem-solving attempts, and plan targeted review sessions. When used this way, the error notebook can support a more reflective, self-directed stance, moving closer to Resnick’s image of learners iterating on

their own projects and ideas over time (Resnick, 2014). Whether students use this feature mainly for repetitive practice or as a way to reflect on their own learning journey depends less on how the software is programmed and more on how teachers, parents, and students choose to use it in everyday study routines.

Doubao Aixue's picture-book creation feature stands out as a more constructionist element. In this mode, the learner contributes the initial idea or theme for a story, and the AI collaborates by generating narrative text and images. Instead of converging on a single correct answer, the interaction centers on creating a unique artifact that can be revised, extended, or shared. This resonates with Resnick's four P's: students work on a project (a storybook), often grounded in personal interests or passions, and they have opportunities to play with different narrative and visual forms (Resnick, 2014). Although the current implementation may not fully realize the "peers" dimension, for example, through public sharing and remixing, the core structure positions the learner as creator with agency rather than a problem-solver following a fixed template.

Both platforms also include "Learn & Discover"—the feature that answers factual questions posed by learners. In Doubao Aixue, this function allows students to point at or photograph objects and receive explanations, while Gauth includes general discovery and explanation tools alongside its homework functions. At first glance, these tools look strongly instructionist because they transmit declarative knowledge: the system delivers definitions, explanations, or descriptions in response to a query. However, from a constructionist and situated perspective, the origin of the question matters. When the learner's own curiosity or everyday experience triggers the inquiry, for example, asking why a particular animal behaves a certain way or what a specific object is, the interaction begins to resemble the interest-driven exploration that Resnick associates with passion and play (Resnick, 2014), and that Lave links to learning embedded in ongoing activity rather than separated from it (Lave, 1991). In other words, the content is delivered instructionally, but the activity structure can still support personally meaningful sense-making.

The reading summary feature on Gauth condenses a text into key themes, plot elements, and interpretations. Pedagogically, it allows learners to quickly access a synthesized view of a

reading. In an instructionist frame, this can be very efficient for delivering explanations. Gauth's writing assistant and essay-planning tools follow a similar logic. They help students generate outlines, draft essays, and refine language to meet conventional academic expectations.

Across both platforms, AI often acts as a powerful source of explanations, examples, and corrections. This pattern aligns with instructionist traditions that value clarity, efficiency, and reduction of cognitive effort during initial learning. At the same time, several features, especially Doubao's error notebook, picture-book creation, and curiosity-driven discovery tools, create spaces where learners can engage in practices more constructively. But it depends on how students, families, and schools use it.

In summary, Gauth and Doubao Aixue are not purely "instructionist" or purely "constructionist." Instead, they assemble a range of features that lean in different directions. Their dominant pattern is to support explicit instruction, step-by-step modeling, and practice on well-defined tasks. Yet within that pattern, there are meaningful openings for learner-driven creation and reflection. Recognizing this mix helps set up Part III, where I examine how these design choices interact with Justin Reich's ideas about the "curse of the familiar" and the EdTech Matthew Effect in both the U.S. and Chinese contexts.

Part III. The Curse of the Familiar and the EdTech Matthew Effect in Gauth and Doubao Aixue

In *Failure to Disrupt*, Justin Reich (2020) argues that educational technologies rarely transform schooling in the ways their advocates imagine. Instead, they tend to reproduce familiar instructional practices and existing inequalities. This section applies Reich's analytical framework to Gauth and Doubao Aixue, examining two interconnected patterns. First, I show how both platforms exemplify the "curse of the familiar" by automating existing homework and exam-preparation routines rather than disrupting them. Second, I explore how the "EdTech Matthew Effect" shapes who benefits most from these platforms, arguing that students with stronger resources, skills, and support are better positioned to extract deeper learning value from AI tools. Building on the policy and pedagogical analyses in Parts I and II, I demonstrate that although these platforms were designed in different governance systems, they converge in scaling familiar forms of schooling and stratifying access to the most powerful uses of AI.

Reich describes the "curse of the familiar" as the tendency for new tools to be most easily adopted when they fit smoothly into existing routines, incentives, and structures (Reich, 2020). Both Gauth and Doubao exemplify this pattern by deploying their most powerful capabilities (rapid solution generation, instant feedback, large-scale summarization) to support stubbornly persistent structures rather than enabling new forms of learning. Gauth's photo-based problem solving, step-by-step solutions, reading summaries, and writing assistance don't question the familiar sequence of assign, solve, and submit; they simply accelerate it. If homework is already organized around demonstrating that students can reproduce a given method or generate a conventional interpretation, Gauth helps them do so more quickly and accurately. It reinforces the idea that homework's value lies in producing correct answers and polished products efficiently rather than inviting teachers or students to rethink homework as a space for exploration or long-term projects.

Doubao enacts a parallel dynamic within China's centralized, exam-driven system. Its photo Q&A, homework grading, problem explanations, and error notebook directly mimic long-standing exam preparation practices. Many students already keep physical notebooks of mistakes, redo problems, and rely on teachers for step-by-step explanations. Doubao encodes this digitally: mistakes get automatically captured, explanations come from an AI teacher persona, and practice takes the form of repeated engagement with similar items. Even its more open-ended features (picture book creation, curiosity-driven questioning) coexist with and are arguably overshadowed by exam-focused functionalities. Many students may experience real gains in accuracy, confidence, and efficiency, but these gains occur within the gravitational pull of the exam regime rather than pushing against it. Technology "succeeds" by fitting into the familiar, leaving deeper logics of schooling largely unchanged.

Beyond reproducing familiar practices, Reich argues that educational technologies also tend to reproduce and amplify existing inequalities through what he calls the "EdTech Matthew Effect." Drawing on the broader Matthew Effect literature, he suggests that students and schools who already have more resources, skills, or support are best positioned to benefit from new technologies, while those with fewer supports gain less (Reich, 2020). This pattern operates through both economic and pedagogical dimensions in Gauth and Doubao.

For Gauth, the app requires capable devices, a stable internet connection, and paid subscriptions for unlimited and additional services. Students whose families can afford premium tiers receive more extensive support. Premium plans not only remove usage limits but also make it easier to access human expert help at any time, as tutors can solve complex problems when the AI's answer is incomplete or incorrect (Amdad, 2024). Students who cannot pay have fewer opportunities to ask additional questions, receive detailed explanations, or consult human tutors, and must rely more heavily on whatever the algorithm alone can provide.

There is also a cognitive dimension. The core math features in both platforms implement the worked example effect: rather than solving problems from scratch, learners study fully worked solutions and apply the same schema to similar items (Sweller et al., 1998). For novice learners still building basic schemas, this guided support can be highly beneficial because it reduces unproductive search and focuses attention on key steps. However, Sweller also describes an expertise reversal effect: as learners become more experienced, very detailed step-by-step guidance can become less effective or even counterproductive compared to more open problem-solving (Sweller, 2016). Instead of relying on fully worked solutions for every problem, advanced students may benefit more when they use AI to extend their thinking by exploring alternative methods, checking conjectures, or engaging in complex tasks.

This creates a stratified pattern of use. For novice students with limited support, AI-based worked examples may provide crucial scaffolding they otherwise lack, potentially helping them catch up. For more advanced or well-supported students, the same tools can be used in richer ways: as checks on reasoning, starting points for extension problems, or components of larger projects. In practice, students from more advantaged backgrounds are more likely to receive guidance on using AI in these deeper ways, from teachers who integrate tools into inquiry-based lessons or parents who help turn solutions into opportunities for reflection. Students with fewer supports may rely on AI primarily for answer-getting and short-term completion, gaining efficiency but less conceptual flexibility. The same platform can widen gaps if one group learns to use it as a thinking partner while another uses it mainly as an answer engine.

The same logic applies to Doubao. Even if the app is free or widely available, its value depends on how it gets woven into students' everyday study routines. Learners whose families or schools

help them build regular habits with the error notebook, interpret AI explanations, and balance exam preparation with creative features are likely to experience richer learning trajectories. Others may rely mainly on quick solutions with fewer opportunities for deeper reflection.

Taken together, the curse of the familiar and the EdTech Matthew Effect suggest that Gauth and Doubao Aixue are unlikely, by themselves, to transform U.S. or Chinese education. Instead, they mostly reinforce existing structures (homework routines and exam regimes), and their most powerful benefits are likely to flow to students and schools already well-positioned. At the same time, both platforms include small spaces for creative, reflective, and curiosity-driven activity. In the long run, their impact may depend on whether educators, families, and designers choose to build on these less familiar uses, rather than using Gauth and Doubao only as faster ways to complete traditional homework and exam preparation.

Conclusion

This comparative analysis reveals that AI tutoring platforms are not culturally neutral tools that simply get translated for different markets. Gauth and Doubao Aixue demonstrate how the same core technology gets reconstructed through policies, local educational ideologies, and cultural expectations. Yet beneath their surface differences, consumer tool versus policy-compatible tutoring service, both platforms share instructionist pedagogical foundations and reinforce assessment-driven learning. How these platforms actually function varies tremendously based on policy contexts, stakeholders, and communities of use. Chinese parents increasingly use Doubao to help with their children's math homework. U.S. students use Gauth to navigate homework and test prep. The same features can serve as thinking partners or answer engines depending on how teachers, parents, and students integrate them into everyday practices.

Understanding these variabilities matters for researchers studying EdTech, for educators and families making choices about which tools to use and how, and for policymakers trying to shape AI's role in education. The question is not whether AI tutoring will transform or reinforce existing educational structures, but rather how different actors: educators, families, students, policymakers, and designers will shape its integration into diverse educational contexts.

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